

Q1. Load the Titanic dataset (titanic.csv) Create a histogram showing the number of passengers in each passenger class. Create a Pie Chart showing the survival distribution.

[10 Marks]

Q2. Apply Polynomial Regression to predict song popularity using audio features such as danceability, energy, and tempo. Evaluate the model using MAE, RMSE, and R^2 Score, and visualize the fitted curve.

[20 Marks]

OR

Q2. Use the K-Nearest Neighbors (KNN) algorithm to classify emails as spam or non-spam using the Spambase dataset. Preprocess the data, train the KNN classifier, and evaluate its performance using accuracy, confusion matrix, precision, recall, and F1-score.

[20 Marks]

Q3. Viva

[5 Marks]